Project Rationale

1. Statement of the problem :
How does an epidemic spread ? / How to limit the spread of an epidemic ?

2. Objectives :
Represent the spread of an epidemic when different factors come into play (Epidemic rate, vaccination, social distance…)

3. Expected outcome :

- When the epidemic rate is more important the spread should be faster
- If a part of the population is vaccinate the spread should be less important
- If people follow a social distance the epidemic should be under control to a certain extent

4. Justification of the chosen approach :
I choose to use netLogo because it simple and it show well what i want to represent. I try to represent it the most simple way how a realistic epidemic spread works, it's why i use some parameters like the number of initial-infected, the propogation rate, the social distance… A real epidemic is more complicate but i choose to keep the thing simple for having a more comprehensive model.

5. Analysis of the model :

- Evaluation of system performance : I think the model show how an epidemic spread, it is simple but wee can see well the change about the infection when we change the differents parameters.
- Limitations and assumptions : The time needed for recovering from the epidemic is the same for all (in reality it not). All people recover from the epidemic at a time (no one died from it). People who have been vaccinated cannot get sick, same for the people that have been healed.
- Suggestions for improvement and future work : Can add others parameters like the lockdown, wearing a mask. Ensure that the time needed for healing isn't the same for everyone and also add the possibility that the healed people can be sick another times.